

Tone Stacks



Duncanamps

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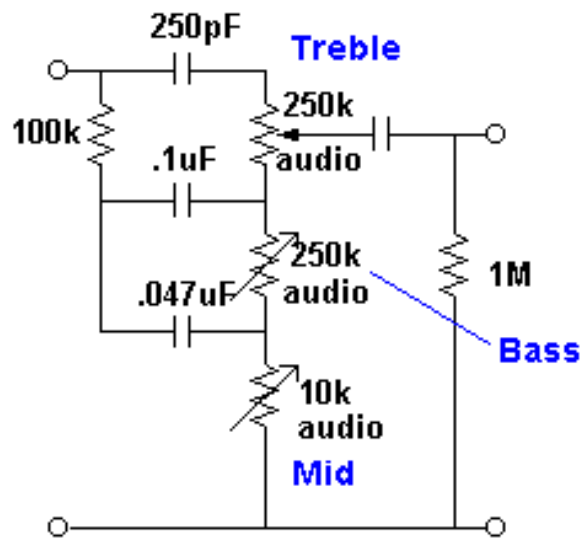
When designing the [DDS-2250V](#), I had the choice of the traditional Fender/Vox/Marshall tone stack using Bass/Mid/Treble controls, or using a passive version of the Baxandall Bass/Treble tone stack which is popular on hi-fi equipment. This is known as the James tone stack.

On the basis that "the more knobs, the better", I started the design using the FVM tone stack. When [SPICE](#) modelling the effects of the stack, it became apparent that the tonal control was limited. Furthermore, the effect of the mid control on everything else was more than noticeable.

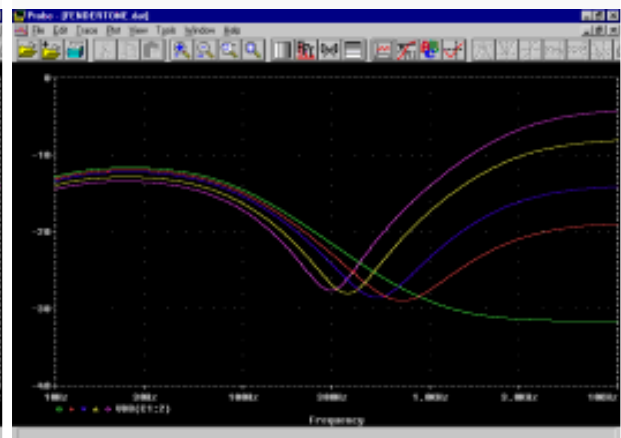
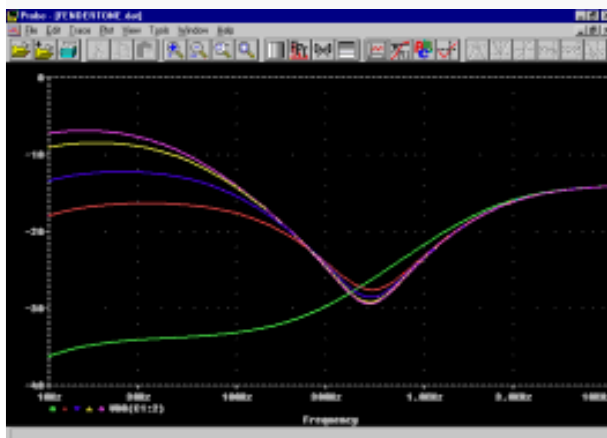
After some rough working out, it became apparent that more different tones were available using James' bass/treble combination than the FVM circuit. Not believing that two knobs gave more variation than three, I double checked everything. Whichever way I looked at it, it had to be James....

Most of the following images can be clicked on to view a larger version approximately three times the size.

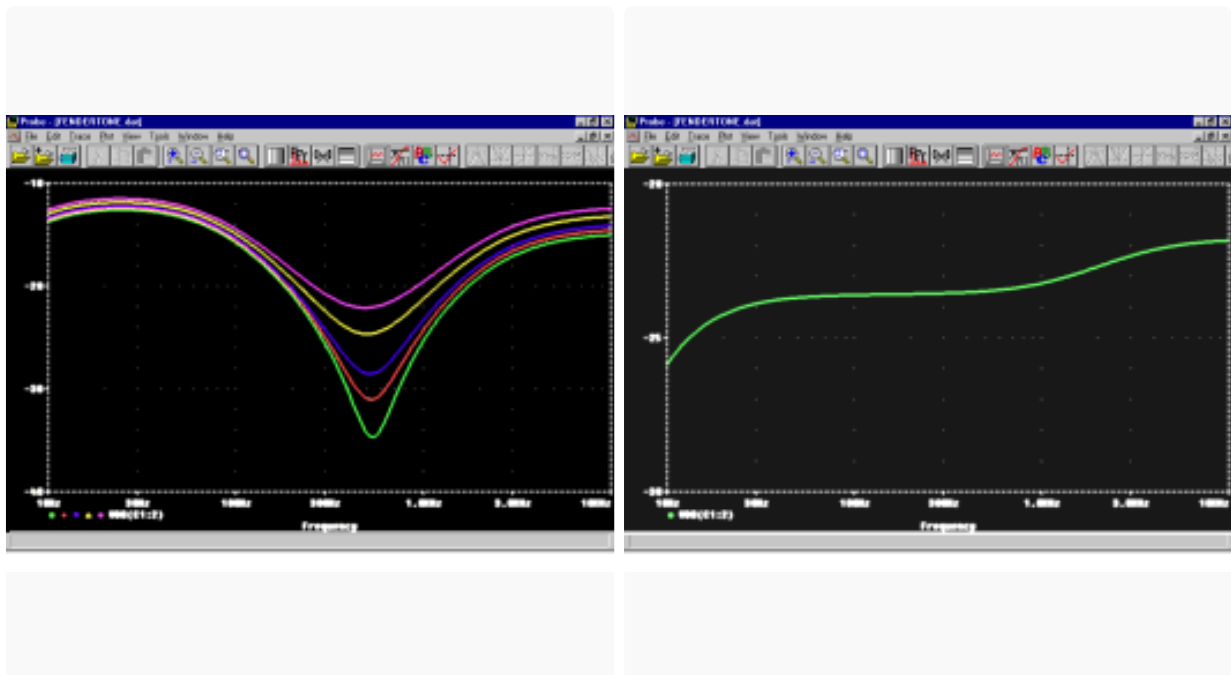
Here's the FVM tone stack, which I've annotated with the control names - this neat example was pulled from Dave Cigna's guitar amp page, and much useful information on operation of the pots and surrounding impedances was provided by Mark Garvin. Indeed, check out Marks excellent post on the [Baxandall tone stack](#) which covers the theory in a lot more detail :



The calculations for the FVM tone stack use audio taper controls with 30% of resistance at midway. Also assumes 40k source impedance and a load of 1 meg (as above). Here's the bass response of the FVM circuit - modelled with PSpice - at 0, 2.5, 5, 7.5, and 10 on the dial. Lots of mid dip. However, at 81Hz (low E), 20dB of variation is possible. The pic on the right is the treble response:



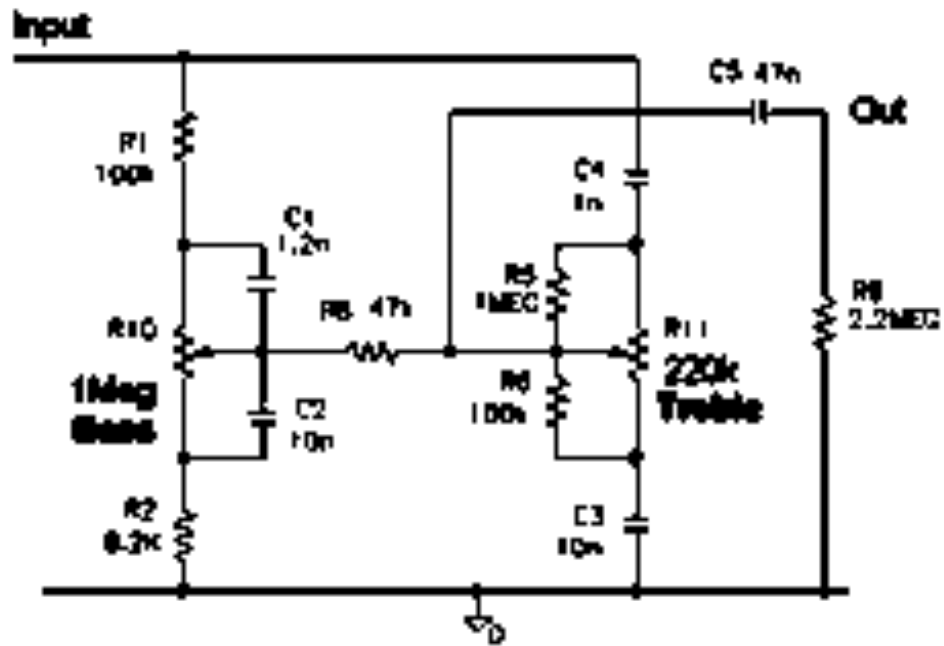
If we take a stand at 6kHz which is about the practical limit for many 12" guitar speakers, a variation of 26dB is possible. Notice how the mid scoop moves... Next, we can take a look at the mid response. Whilst cutting the mid response, it cuts everything else down too. Turning the control between 0 and 5 doesn't do very much. Is it possible to get a mid boost? The diagram on the right shows what happens when you have bass=0, mid=10, treble=0.



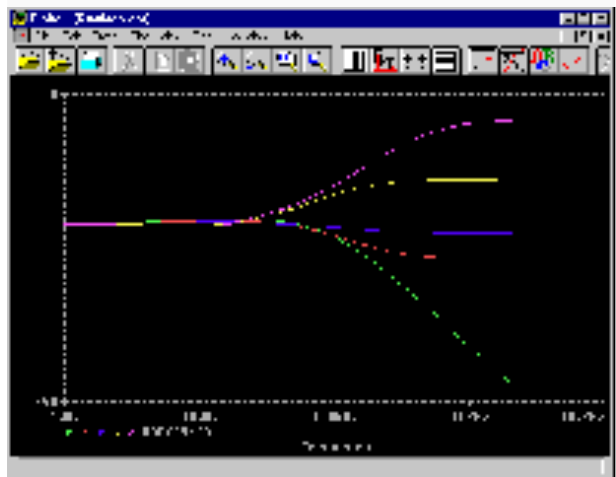
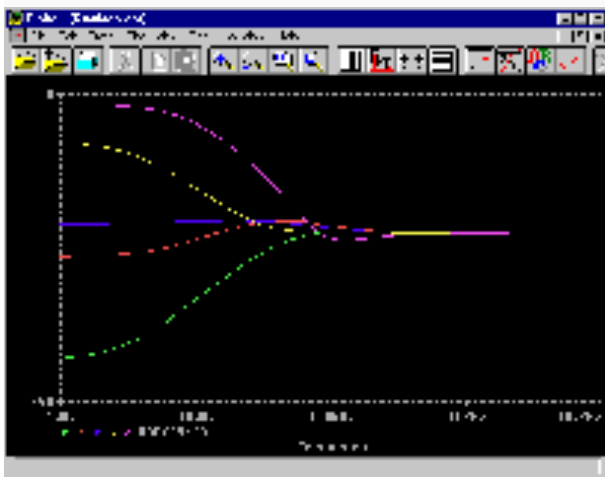
The response above is relatively flat. It becomes apparent then, that the FVM circuit cannot offer any kind of mid boost - no matter how the controls are set.

Looking at the passive James tone stack, it has just two controls in its classic form, bass and treble. For hi-fi use, it is traditional to set the bass turnround point at 100Hz, and the treble at 10kHz. For guitar work, it is more appropriate to halve the treble point due to the lower frequencies involved. The pots in this circuit are audio taper (log), and assume 10% at the halfway position.

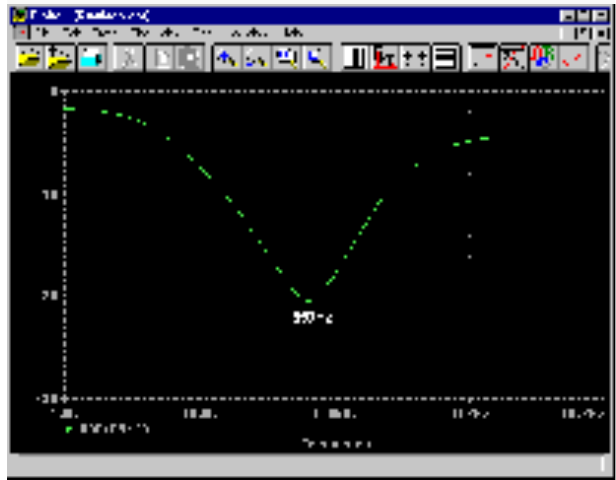
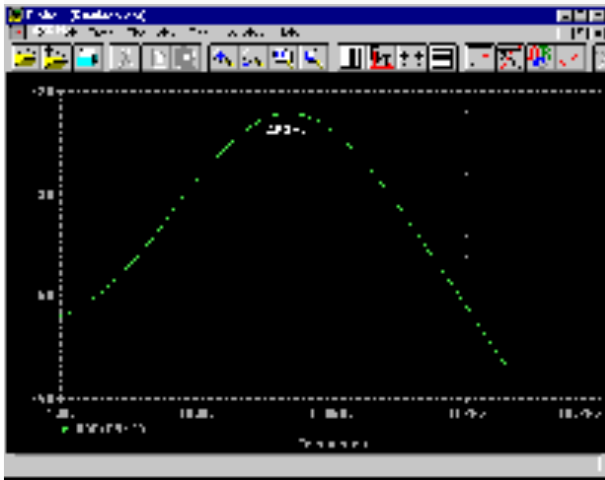
For the DDS-2250V, this stack has been driven from a lower impedance (15k ohms). Load is 2.2 meg. Please note that driving this stack from a higher source impedance (such as straight from the anode of a 12AX7 via a coupling capacitor) will not give the same results. Alternative values for a 40k source impedance and a 1meg load would be change $C1=C4=470p$, $C2=C3=4.7n$, and change $R2$ to 18k.



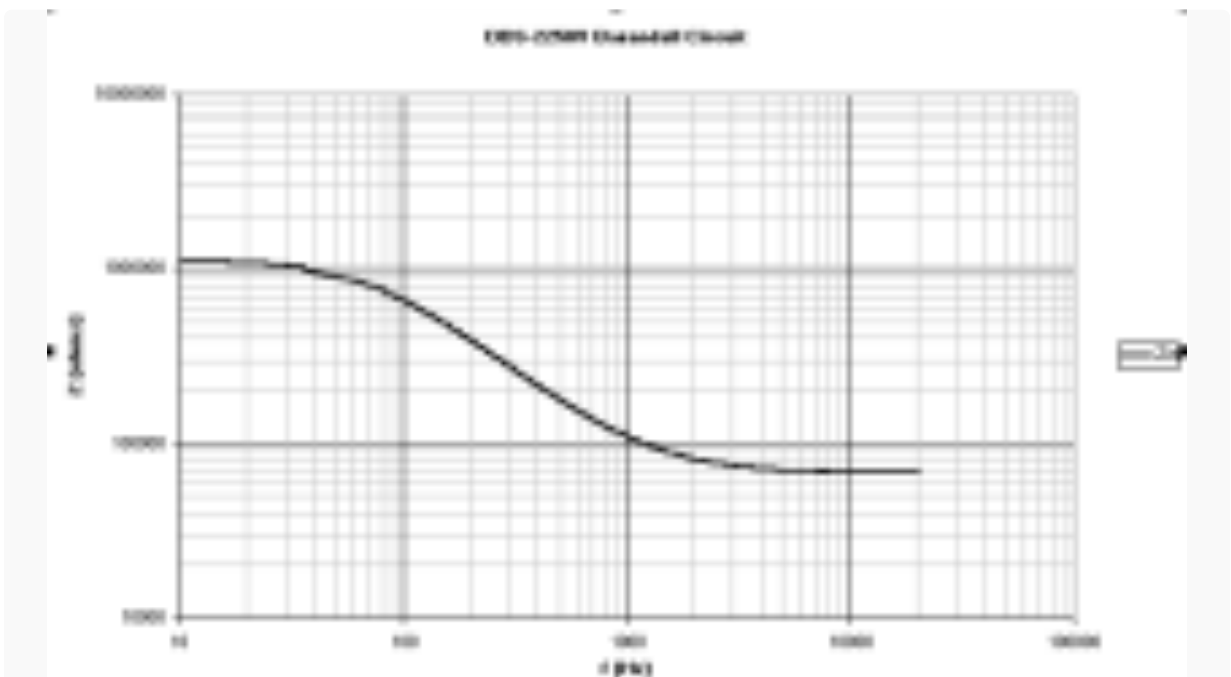
Check the following performance charts against the FVM circuit. First moving the bass control through 0, 2.5, 5, 7.5, and 10 with treble at 5. On the right, moving the treble control through its range whilst keeping the bass at 5:



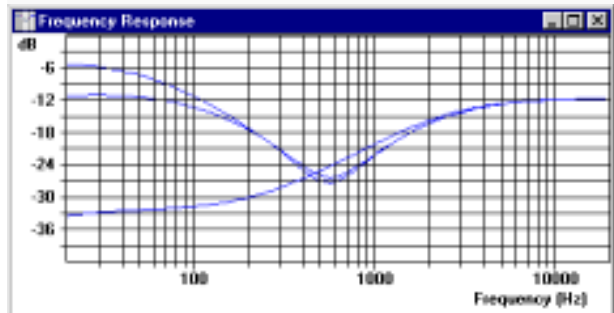
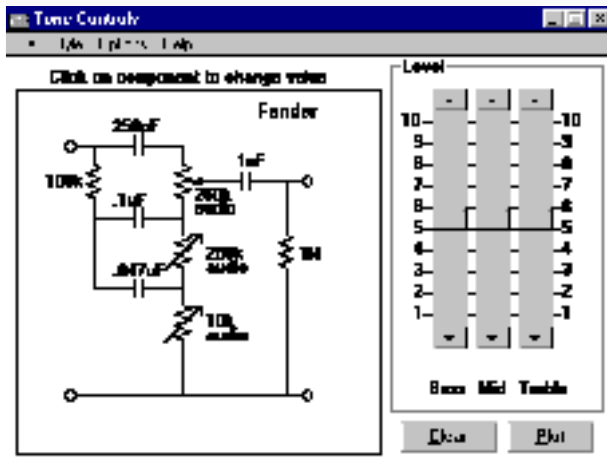
Mid boost, achieved by setting both controls to zero (gain will obviously have to be increased), and mid scoop obtained by turning both controls up to 10:



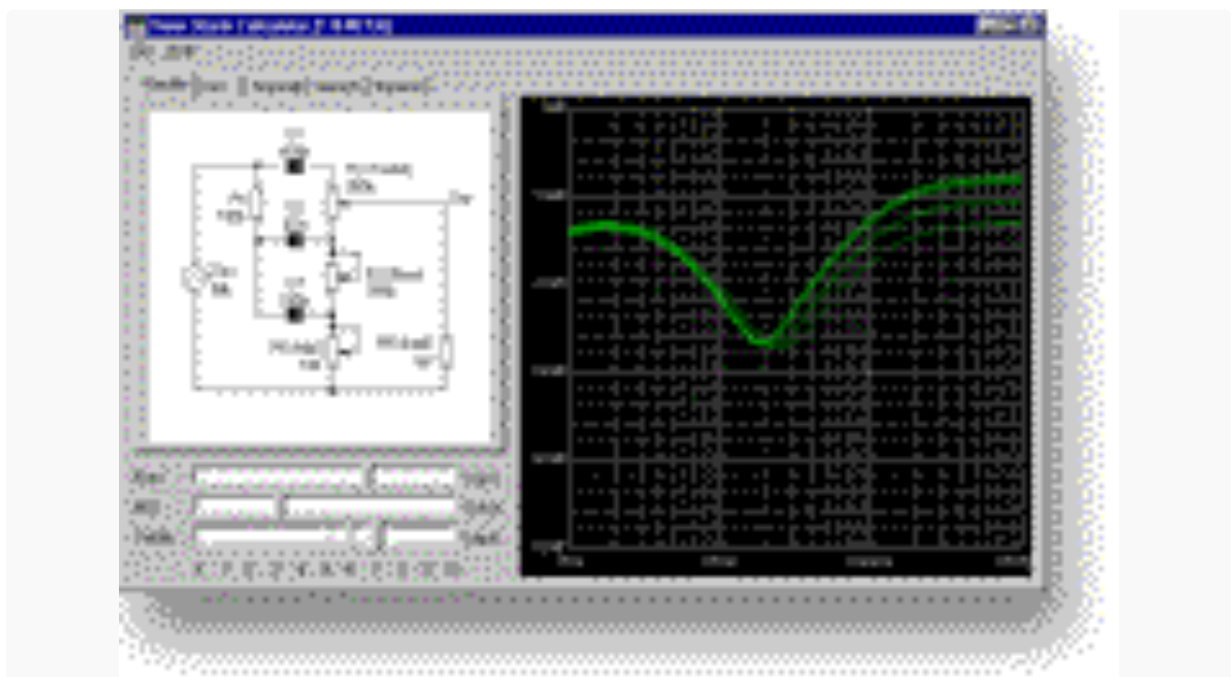
One question that has cropped up a couple of times now, is what is the load impedance of the James stack? Heres a picture!! Don't forget you can click on it to see a bigger version...



This is a very cool tool which will allow **Fender**, **Vox**, and **Marshall** tonestacks to be modelled.



The use interface is simple, and you can build up a series of plots in the plot window to check the effect of altering the controls. On the subject of controls, these can be audio or linear taper and the values can be altered, as can any of the other component values.



Note: Dave's site is no longer operational. However, you can download a zipped version of his Tone Control Simulator by [clicking here](#).

I've written one along similar lines (pictured on the right), and you can [click here](#) to download a copy of it.